DATA 119 Group Project Proposal

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Dataset Description:

Clash Royale is a competitive multiplayer video game where two players seek to destroy their opponent's "towers" using their own troops, summoned by playing cards in real-time. Before a match, players must build an 8-card deck out of a pool of over 120 unique cards. These 8 cards, each summoning a unique unit, often determine the player's success.

There are two types of data within the datasets we are using: decks and cards. Deck data aggregates all the matches in the past 7 days (around 800,000 matches in most weeks) and shows the most used or successful decks. The 7 days allow the data to reflect the current meta of the game. Deck statistics include average elixir (the cost to play each card), the sum of the four lowest elixir cards, usage rate, win rate, draw rate, loss rate, and rating, which is calculated from usage rate and win rate. Card data also aggregates all the matches in the past 7 days, with similar statistics like usage rate and win rate, but also have statistics to characterize the cards, such as elixir cost, rarity, card type, mobility, target, hitpoints, damage, etc.

Project Goals:

There are many questions that can be asked when it comes to deck-building in Clash Royale. What makes a good deck? Are there different types of decks? How accurately can the outcome of a matchup be predicted based solely on the two decks? We aim to answer these questions by analyzing the relationships between card and deck characteristics and their win rates, arriving at a prediction of the success rate of a given deck.

There are also a number of questions related to deckbuilding. If given several cards, is there a statistically optimal way to complete the rest of the deck? How can a player randomly generate a good deck? How do card synergies work? How do special abilities such as evolution or champion cards affect decks? Using the relationships discussed above, we hope to produce a deckbuilding method based on patterns found in the best-performing decks.

Data Provenance:

The data comes from a variety of sources. The official developers of the game have an API called Clash Royale API for their game's statistics, and there are also other databases such as RoyaleAPI and Stats Royale that provide different statistics. Any additional information about the cards themselves can also be found in-game, since there are only around 120 unique cards.